

Auteurs: Mathijs Pittoors
 Studierichting: Informatica
 Oefeningenreeks 2E nr. 26.38

$$4^{\log_x 3} - 4 = 16^{\log_x \sqrt[4]{3}} + 2^{\log_x(3x)}$$

$$\Leftrightarrow 2^{2\log_x 3} - 4 = 2^{4\log_x \sqrt[4]{3}} + 2^{\log_x 3 + \log_x x}$$

$$\Leftrightarrow 2^{2\log_x 3} - 4 = 2^{2\log_x 3} + 2 \cdot 2^{\log_x 3}$$

Neem $y = 2^{2\log_x 3}$

$$\Leftrightarrow y^2 - 4 = y + 2y$$

$$\Leftrightarrow y^2 - 3y - 4 = 0$$

$$D = b^2 - 4ac = 25$$

$$\Leftrightarrow y = \frac{3 \pm 5}{2}$$

$$\Leftrightarrow y = 4 \text{ of } y = -1$$

Opnieuw invullen:

$$-1 = 2^{2\log_x 3}: \text{ Geen oplossingen in } \mathbb{R}$$

of

$$4 = 2^{2\log_x 3}$$

$$\Leftrightarrow 2^2 = 2^{2\log_x 3}$$

$$\Leftrightarrow 2 = 2\log_x 3$$

$$\Leftrightarrow x = \sqrt{3}$$

References